

Model Validation Report: Hull–White / G2++ Bermudan Swaption Engine

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Abstract

This report validates a Monte Carlo Bermudan swaption engine built on the one-factor Hull–White and two-factor G2++ Gaussian short-rate models. Validation follows three independent axes: (i) cross-validation of prices against the QuantLib library (v1.42.1) using matched conventions; (ii) self-consistent numerical evidence (Monte Carlo convergence and an Andersen–Broadie primal–dual policy bound); and (iii) grounding of all model inputs in real, reproducible market data from the Federal Reserve Economic Database (FRED). Every figure quoted below is the output of a script in `research/`, runnable against public data. Limitations are stated explicitly, including those that cannot be closed without a licensed swaption-volatility surface. Software versions: QuantLib 1.42.1, NumPy 2.2.6, SciPy 1.15.3.

1 Models

The one-factor Hull–White short rate follows the risk-neutral process

$$dr_t = (\theta(t) - a r_t) dt + \sigma dW_t,$$

with $\theta(t)$ chosen so the model reproduces the initial discount curve exactly. The two-factor extension (G2++, Brigo–Mercurio) writes $r_t = x_t + y_t + \varphi(t)$ with

$$dx_t = -a x_t dt + \sigma dW_t^{(1)}, \quad dy_t = -b y_t dt + \eta dW_t^{(2)}, \quad dW_t^{(1)} dW_t^{(2)} = \rho dt,$$

and a deterministic shift $\varphi(t)$ fitting the curve. Both are Gaussian, hence consistent with the market’s post-2014 convention of quoting swaption volatility as normal (basis-point) vol, and both admit negative rates by construction.

2 Numerical method

Early exercise is solved by Longstaff–Schwartz least-squares Monte Carlo: forward simulation of the short rate, then backward induction estimating the continuation value by regression on a weighted-Laguerre basis with a ridge-regularised normal-equations solve. Quasi-Monte Carlo draws come from a scrambled Sobol sequence mapped through the inverse normal CDF. Discounting uses the simulated money-market account.

3 Validation results

3.1 European reconciliation (isolates conventions)

On a flat 3% curve with $a = 0.05$, $\sigma = 0.01$, a 2Y-into-3Y European payer swaption:

Quantity	Engine	QuantLib	Difference
ATM forward rate	3.0455%	3.0455%	exact
Jamshidian European price	1.369243×10^{-2}	1.369977×10^{-2}	0.07 bp of notional

The 0.07 bp residual is the day-count difference (integer-year times vs ACT/365). Conventions are reconciled.

3.2 Bermudan vs QuantLib tree, and Monte Carlo convergence

Same instrument and model, three exercise dates. QuantLib trinomial tree (200 steps) = 1.576184×10^{-2} .

QMC paths	LSMC price	95% CI half-width	vs tree
1,024	1.573579×10^{-2}	11.6 bp	−0.26 bp
4,096	1.574304×10^{-2}	5.8 bp	−0.19 bp
16,384	1.573035×10^{-2}	2.9 bp	−0.31 bp
65,536	1.572892×10^{-2}	1.45 bp	−0.33 bp

Two independent methods (Monte Carlo LSMC vs trinomial tree) agree to within Monte Carlo error. The confidence interval scales as $1/\sqrt{N}$: 64× the paths gives exactly 8× tighter CI. The LSMC sits a few tenths of a bp below the tree, the sign of the known Longstaff–Schwartz low bias.

3.3 Policy optimality: Andersen–Broadie primal–dual bound

The dual upper bound built from the policy’s own 4-term basis gave a gap of ~ 8.3 bp that did *not* shrink with more inner simulations (8.3 bp at 200 inner paths, 8.3 bp at 5,000), proving the gap is value-surface approximation error, not Monte Carlo noise. Replacing the value surface with a degree-5 regression collapsed the gap to ~ 0.04 bp. The duality gap is therefore statistically consistent with zero (well inside the lower-bound CI), which **proves the LSMC exercise policy is near-optimal** — a stronger statement than the single-exercise Jamshidian check, which validates only the terminal boundary condition.

3.4 Two-factor (G2++) validation

Empirical motivation. Principal component analysis of daily changes in the US Treasury curve (FRED DGS2, DGS5, DGS10, DGS30, 1,365 trading days, 2021-01-04 to 2026-06-17):

PC	Variance	Cumulative	Interpretation
1	87.12%	87.12%	level (loadings 0.51, 0.56, 0.51, 0.40)
2	11.16%	98.28%	slope (0.67, 0.15, −0.32, −0.66)
3	1.39%	99.67%	curvature

A one-factor model captures only the level factor and *forces all rates to be perfectly correlated*, discarding the 11% slope factor.

Implementation. The G2++ bond reconstruction reproduces the initial curve to machine precision ($P(0, T; 0, 0)$ vs market: error 0.00×10^0).

Decorrelation. Terminal correlation of zero rates at a 1-year horizon (G2++ params $a = 0.5$, $b = 0.05$, $\sigma = 80$ bp, $\eta = 60$ bp, $\rho = -0.7$):

corr(2Y, 30Y)	Value
One-factor Hull–White	1.00 (structural)
G2++	0.77
Empirical (real FRED)	0.57

One-factor is forced to 1.00, contradicted by the data (0.57). G2++ reproduces the correct decorrelation shape; the illustrative parameters under-decorrelate versus reality, which calibration would close.

Pricing cross-validation. A 2Y-into-3Y G2++ European payer swaption:

Method	Price
Engine Monte Carlo	6.176933×10^{-3} (± 0.39 bp)
QuantLib FdG2 engine	6.187076×10^{-3}
Difference	0.10 bp (within MC error)

The two-factor engine is validated end-to-end against an independent finite-difference implementation.

3.5 Risk sensitivities by algorithmic differentiation (AAD)

Greeks are computed by reverse-mode automatic differentiation of a policy-fixed pricer: holding the exercise policy fixed (justified to first order by the envelope theorem), the price is a smooth function of the curve nodes and σ , and a single reverse pass returns the entire gradient (all key-rate deltas and vega). Against bump-and-revalue on the same pricer the agreement is to $\sim 10^{-6}$ (vega \$150.32/bp by both methods; 2Y key-rate DV01 $-\$68.78$ by both; 5Y $+\$239.97$ by both).

The cost advantage follows the cheap-gradient principle: one reverse pass costs a constant $\sim 3.3\times$ a single valuation *independent of the number of inputs*, while bump-and-revalue scales linearly.

Risk factors	1 valuation	AAD reverse	Bump-all	AAD speedup
14	5.3 ms	20.9 ms	73 ms	$3.5\times$
42	5.3 ms	16.6 ms	225 ms	$13.6\times$
72	5.4 ms	18.0 ms	377 ms	$21.0\times$
132	5.3 ms	17.7 ms	715 ms	$40.3\times$

The speedup grows with the size of the risk ladder; a realistic book (key-rate buckets plus a vega ladder) exceeds 100 factors.

4 Real-market parameterisation

4.1 Data sources (reproducible)

All market data is public and reproducible from FRED: overnight SOFR and SOFR30/90/180DAYAVG; H.15 Treasury constant-maturity yields DGS1... DGS30. The discount curve used for pricing is dated **2026-06-17** (13 nodes, overnight 3.63% to 30Y 4.93%). Curve-implied par swap rates are bootstrapped from this curve and labelled as model-implied, not market mid quotes (FRED discontinued its swap-rate series).

4.2 Volatility estimated from history, not assumed

Fitting the short-rate process to the full daily SOFR series (2,050 observations, 2018-04-03 to 2026-06-17):

σ (bp/yr)	Window	Note
169.4	full history	contaminated by the Sept-2019 repo spike (SOFR 5.40%) and 2022–23 hikes
75.1	last ~ 2 y	trustworthy estimate (± 2.4 bp, 95%)
72.6	last ~ 1 y	regime-dependent

The realized-vol and Ornstein–Uhlenbeck-MLE estimates of σ agree to 0.1 bp, an internal validation of the estimation. Mean reversion a is *not* reliably identifiable from a non-stationary level series (OU MLE: 0.37 ± 0.31), consistent with a being a calibration parameter in practice.

Price impact. Repricing the 2026-06-17 Bermudan on the real curve:

σ source	Bermudan price
assumed 100 bp	\$15,292.68
realized ~ 2 y, 75.1 bp	\$11,495.95
realized ~ 1 y, 72.6 bp	\$11,115.16

Moving from an assumed to a sourced σ changed the price by $\sim 25\%$, demonstrating that the parameter was material, not cosmetic. **Caveat (measure):** diffusion volatility is measure-invariant, so realized vol is a legitimate estimate of the true σ ; however, market swaption prices embed *implied* vol, which exceeds realized by a variance risk premium, so the realized-vol price is a physical/historical anchor below the market-consistent price. Only an implied-vol calibration closes that gap.

5 Risk analytics (real curve, 2026-06-17)

2Y-into-3Y ATM payer Bermudan, \$1,000,000 notional, computed with common random numbers for stability.

- **Parallel DV01:** \$125.71 / bp.
- **Key-rate DV01:** +\$240.73 at 5Y, $-\$56.10$ at 2Y, $-\$57.52$ at 3Y, negligible beyond. Long the back, short the front, as expected for a payer.
- **Model vega:** \$152.38 per 1 bp of σ .
- **Exercise distribution:** 37.3% at 2Y, 14.4% at 3Y, 10.7% at 4Y, 37.7% never ($P(\text{ever}) = 62.3\%$).
- **Exposure (time- t mark-to-market):** EPE \$19.1k / \$4.0k / \$0.7k and PFE(97.5%) \$70.3k / \$28.8k / \$10.1k across the three exercise dates, the expected run-off profile.

Honest finding: the key-rate buckets summed to \$130.98 versus a \$125.71 parallel DV01 ($\sim 4\%$ gap). The cause is named: the curve uses natural cubic-spline interpolation, which is non-local, so single-node bumps do not cleanly partition a parallel shift. A production setup would use local interpolation or a full Jacobian for bucketed risk.

6 Conventions and limitations

Convention alignment (post-LIBOR). USD LIBOR ceased in June 2023; swaps reference SOFR, a risk-free overnight rate with no tenor basis, so the single-curve approach is more defensible now than in the LIBOR era. The Gaussian model is consistent with normal-vol

quoting and admits negative rates. The single SOFR curve is consistent with SOFR-CSA collateral discounting.

Stated limitations.

1. **Single-factor smile.** A Gaussian model (1F or G2++) structurally cannot produce a volatility smile/skew. Capturing it requires a different model class (SABR, Cheyette-with-skew). G2++ addresses curve decorrelation, not smile.
2. **Implied-vol calibration is absent.** Parameters are historically estimated (σ) or conventional (a), not calibrated to a swaption-vol surface, because no free, verifiable surface exists. The production standard for Bermudans is calibration to co-terminal European swaptions. This is the single largest gap and is deliberately left open rather than filled with fabricated quotes.
3. **Market data scope.** Multi-curve OIS/projection separation, a vol cube, cash- vs physical-settled conventions, collateral/CSA and funding (FVA), full calendars and day-count are not modelled; swap conventions are stylised.
4. **Key-rate non-locality** under cubic-spline interpolation (Section 6).
5. **Governance.** Reproducibility (pinned seeds, dependencies, dated data snapshots) and validation records (this report and **research/**) are in place; observability, deterministic builds, and an independent validation sign-off are not, so “production-grade” is not yet claimed.

This evidence set — independent benchmarking against QuantLib, convergence analysis, primal–dual policy validation, real-data parameterisation, and documented limitations — is the form of evidence expected under model-risk guidance such as the Federal Reserve’s SR 11-7.

References

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